House price prediction [Classification]

Case study-

A real estate company wants to predict house prices based on factors like area, number of bedrooms, location, age of the house, and amenities.

Perform the following:

i) Build a suitable ML model

ii) Identify input and output variables

iii) Explain preprocessing steps required

Answer-

i) Build a Suitable Machine Learning Model (3–4 Marks)

Since house price is a **continuous value**, ideally this is a **Regression problem**.

However, if we **convert prices into categories** (e.g., Low, Medium, High), then we can use a **Decision Tree Classifier**.

Chosen Model: Decision Tree Classifier

- The model splits data based on conditions like:
  - Area > 1500 sq.ft
  - Location = Urban or Rural
  - Bedrooms > 3
- Each branch leads to a **price category (Low/Medium/High)**

Need of Decision Tree Classifier

- Easy to understand and interpret
- Handles both numerical and categorical data
- Captures non-linear relationships

Sample Data set-

Area	Bedrooms	Location	Age	Price category
1000	2	1	10	0
1500	3	2	25	1
2000	3	2	23	1
2500	4	3	2	2
3000	5	3	3	2

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier

# Sample dataset
data = {
    'area': [1000, 1500, 2000, 2500, 3000],
    'bedrooms': [2, 3, 3, 4, 5],
    'location': [1, 2, 2, 3, 3],  # encoded
    'age': [10, 25, 23, 2, 3],
    'price_category': [0, 1, 1, 2, 2]  # Low=0, Medium=1, High=2
}

df = pd.DataFrame(data)

# Input and Output
X = df[['area', 'bedrooms', 'location', 'age']]
y = df['price_category']

# Split data
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)

# Model
model = DecisionTreeClassifier()
model.fit(X_train, y_train)

# Prediction
y_pred = model.predict(X_test)
print(y_pred)
```

ii) Identify Input and Output Variables (2–3 Marks)

Independent Variables (Input – X):

- Area of house

- Number of bedrooms
- Location
- Age of house
- Amenities (optional)

Dependent Variable (Output – Y):

- House Price Category (Low / Medium / High)

These features directly influence property value.

iii) Data Preprocessing Steps (Detailed Explanation – 4–5 Marks)

Data preprocessing is a crucial step in House Price Prediction because real-world housing datasets are often **incomplete, inconsistent, and noisy**. Proper preprocessing ensures that the data becomes **clean, structured, and suitable** for building an accurate machine learning model.

1. Handling Missing Values

In real estate datasets, some values may be missing due to unrecorded information or data collection errors. For example, details like **area, age of house, or amenities** might not be available for all entries.

- Missing values can be handled in two ways:
  - **Removing incomplete rows** when a large portion of data is missing
  - **Filling (imputing) values** using statistical techniques
- For numerical features like area or age:
  - Use **mean or median** values

Example:
If the area of a house is missing, it can be replaced with the **average area** of all houses in the dataset.

This step prevents errors and ensures that the model receives complete input data.

2. Encoding Categorical Data

Machine learning models cannot directly process categorical (text-based) data such as location or amenities. Therefore, these values must be converted into numerical form.

- **Label Encoding:** Assigns numbers to categories
  - Example: Urban = 1, Semi-Urban = 2, Rural = 3
- **Binary Encoding:** For Yes/No values
  - Example: Amenities → Yes = 1, No = 0

Example:
Location (Urban, Semi-Urban, Rural) is converted into numbers so that the model can interpret it mathematically.

This step ensures that all features are in a format suitable for model training.

3. Feature Scaling

Different features in house data may have very different ranges. For example:

- Area → 500 to 5000 sq.ft
- Age → 1 to 50 years

If not scaled, features with larger values (like area) may dominate the model and affect learning.

- Feature scaling brings all values to a **similar range**
- Common methods:
    - **Normalization (0 to 1 range)**
    - **Standardization (mean = 0, standard deviation = 1)**

This improves:

- Model **performance and accuracy**
- **Speed of learning (faster convergence)**

4. Removing Outliers

Outliers are extreme values that differ significantly from normal data patterns. These can negatively impact the model.

- Outliers can be identified using:
    - Statistical methods
    - Graphical methods (like box plots)

Example:
A luxury house with an extremely high price compared to others may distort the model's predictions.

Such values can be:

- Removed, or
- Treated separately

This step ensures that the model learns **general trends instead of rare cases**.

5. Train-Test Split

After preprocessing, the dataset is divided into two parts:

- **Training Set (80%)** → Used to train the model
- **Testing Set (20%)** → Used to evaluate the model

This is important because:

- It tests the model on **unseen data**
- Helps measure **accuracy and generalization ability**
- Prevents **overfitting** (model memorizing data instead of learning)

Conclusion

- Decision Tree Classifier can classify house prices into categories
- Preprocessing improves model accuracy
- Model helps real estate companies in pricing and decision-making